

Exploratory Data Analysis (EDA) Summary Report

1. Introduction

The purpose of this report is to perform Exploratory Data Analysis (EDA) on the Delinquency Prediction Dataset. The goal is to understand the structure of the dataset, identify missing values and data quality issues, analyze relationships among variables, detect potential risk indicators related to customer delinquency, and prepare the data for predictive modeling.

2. Dataset Overview

This section summarizes the dataset, including the number of records, key variables, and data types. It also highlights any anomalies, duplicates, or inconsistencies observed during the initial review.

Key Dataset Attributes

- Number of records: 500**
- Key variables:**
 - Age – Customer age**
 - Income – Annual customer income**
 - Credit_Score – Customer credit score**
 - Credit_Utilization – Credit utilization percentage**

- **Missed_Payments** – Number of missed payments
- **Loan_Balance** – Outstanding loan balance
- **Debt_to_Income_Ratio** – Debt-to-income ratio
- **Employment_Status** – Employment category
- **Credit_Card_Type** – Credit card category
- **Delinquent_Account** – Target variable indicating delinquency status
- **Data types:**
 - **Numerical:** Age, Income, Credit_Score, Credit_Utilization, Missed_Payments, Loan_Balance, Debt_to_Income_Ratio, Account_Tenure, Delinquent_Account
 - **Categorical:** Customer_ID, Employment_Status, Credit_Card_Type, Location, Month_1 to Month_6

Initial Review Findings

- **Duplicate records found: 0**
- **Inconsistent values observed in Employment_Status column:**
 - **EMP**
 - **Employed**
 - **employed**

These values should be standardized before model development.

3. Missing Data Analysis

Identifying and addressing missing data is critical to ensuring model accuracy. This section outlines missing values in the dataset, the approach taken to handle them, and justifications for the chosen method.

Key Missing Data Findings

Variables with missing values:

- Income: 39 missing values**
- Credit_Score: 2 missing values**
- Loan_Balance: 29 missing values**

Missing Data Treatment

Recommended treatment:

- Income → Median Imputation**
- Credit_Score → Median Imputation**
- Loan_Balance → Median Imputation**

Median imputation is recommended because it is less sensitive to extreme values and helps preserve the overall distribution of the data.

4. Key Findings and Risk Indicators

This section identifies trends and patterns that may indicate risk factors for delinquency. Feature relationships and statistical correlations are explored to uncover insights relevant to predictive modeling.

Key Findings

Correlations observed between key variables and Delinquent_Account:

- Income: 0.045**
- Credit_Score: 0.035**
- Debt_to_Income_Ratio: 0.034**
- Credit_Utilization: 0.034**
- Age: 0.023**

Findings:

- No strong correlation was observed between any individual feature and delinquency.**
- Income showed the highest positive correlation with delinquency, although the relationship is very weak.**
- Delinquency appears to depend on multiple factors rather than a single variable.**
- Approximately 16% of customers were classified as delinquent accounts.**

Unexpected Anomalies

- **Missed_Payments** showed a weak negative correlation with delinquency, which may require further investigation.
 - **Employment_Status** contains inconsistent category labels that should be standardized.
 - No duplicate records were identified.
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5. AI & GenAI Usage

Generative AI tools were used to summarize the dataset, suggest missing data treatment methods, and assist in identifying patterns within the data.

AI-Generated Insights

- Identification of missing values and data quality issues.
- Interpretation of correlation results.
- Generation of summary findings and recommendations.

Example AI Prompts Used

- "Summarize key patterns in the dataset and identify anomalies."
- "Suggest an imputation strategy for missing income values based on industry best practices."
- "Interpret correlation results for delinquency prediction."

6. Conclusion & Next Steps

The dataset contains 500 customer records and 19 variables related to financial behavior and delinquency risk. Missing values were identified in Income, Credit_Score, and Loan_Balance, while no duplicate records were found. Correlation analysis indicated that no single variable strongly predicts delinquency, suggesting that predictive performance will depend on a combination of features.

Recommended Next Steps

- Standardize inconsistent categorical values.**
- Impute missing values using median imputation.**
- Encode categorical variables for machine learning.**
- Perform feature scaling where required.**
- Address class imbalance in the target variable.**
- Develop and evaluate predictive models such as Logistic Regression, Random Forest, and XGBoost.**
- Analyze feature importance to identify key delinquency drivers.**